

5	(a) (uniform magnetic) flux normal to long (straight) wire carrying a current of 1 A (creates) force per unit length of 1 N m^{-1}	M1 A1	[2]
	(b) (i) sketch: concentric circles increasing separation (<i>must show more than 3 circles</i>) correct direction (anticlockwise, looking down)	M1 A1 B1	[3]
	(ii) $B = (4\pi \times 10^{-7} \times 6.3) / (2\pi \times 4.5 \times 10^{-2})$ $= 2.8 \times 10^{-5} \text{ T}$	C1 A1	[2]
	(iii) $F = BIL (\sin \theta)$ $= 2.8 \times 10^{-5} \times 9.3 \times 1$ $F/L = 2.6 \times 10^{-4} \text{ N m}^{-1}$	C1 A1	[2]
	(c) force per unit length depends on product $I_X I_Y$ / by Newton's third law / action and reaction are equal and opposite so same for both	M1 A1	[2]

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